

Pointwise Truth-RMSE Delta Decomposition for Local versus Global Auto Chart Dimension

Source path: ~/current_projects/geosmooth/split_handoffs/
lps_local_auto_pointwise_decomposition_2026-06-05/
lps_local_auto_pointwise_truth_rmse_delta_decomposition.tex

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Purpose and Main Question

This report records a diagnostic framework for understanding why the local chart-dimension version of the local polynomial smoother (LPS) differs from the global chart-dimension version. The immediate empirical question is not only whether

`chart.dim = "local.auto"`

has lower Truth RMSE than

`chart.dim = "auto".`

The deeper question is *which data points and which local-dimension decisions drive the dataset-level difference.*

The paired HTML report showed the overall Truth-RMSE difference between the two rules. Such a number is useful, but it is too compressed. A small global difference could mean that nearly every point changed a little, or it could mean that a few high-impact points changed a lot while most points were irrelevant. It could also mean that local dimension one helped in one region while local dimension three hurt in another. This report defines five complementary meanings of a pointwise or grouped “component” of the Truth-RMSE delta, then applies them to the first dataset in the current HTML report:

FB01. LA-D1-RAW-N500.

Executive summary for FB01. For this dataset, the global-auto fit used chart dimension 2 everywhere, selected support size 35, Gaussian kernel, and had Truth RMSE 0.046726. The local-auto fit selected support size 34, Gaussian kernel, and used pointwise local chart dimensions 1, 2, 3 with counts 40, 410, 50, respectively. Its Truth RMSE was 0.049115. Therefore local-auto was worse by

0.0023897.

The exact pointwise decomposition below sums to

0.0023897,

which matches the observed Truth-RMSE delta up to displayed rounding.

1 Definitions

Let n be the number of points in a synthetic dataset. At point i , let f_i be the true signal and y_i be the noisy observation:

$$y_i = f_i + \varepsilon_i, \quad i = 1, \dots, n.$$

Let $\hat{f}_i^a$ be the fitted value from the global-auto chart dimension rule and let $\hat{f}_i^{la}$ be the fitted value from the local-auto chart dimension rule. The corresponding truth errors are

$$e_i^a = \hat{f}_i^a - f_i, \quad e_i^{la} = \hat{f}_i^{la} - f_i.$$

The two Truth RMSE values are

$$R_a = \left\{ \frac{1}{n} \sum_{i=1}^n (e_i^a)^2 \right\}^{1/2}, \quad R_{la} = \left\{ \frac{1}{n} \sum_{i=1}^n (e_i^{la})^2 \right\}^{1/2}.$$

The dataset-level Truth-RMSE delta is

$$\Delta_{\text{RMSE}} = R_{la} - R_a.$$

Positive values mean that local-auto is worse; negative values mean that local-auto is better.

Let d_i^{la} be the local-auto chart dimension estimated at point i , and let d^a be the single global-auto dimension. The dimension deviation is

$$q_i = d_i^{la} - d^a.$$

In FB01, $d^a = 2$, while d_i^{la} takes values 1, 2, and 3.

2 Five Ways to Interpret a Pointwise or Grouped Component

The word “component” can mean several related but different things. The following five definitions are deliberately complementary. None by itself answers every question; together they separate algebraic contribution, pointwise error geometry, spatial localization, concentration of influence, and group-level interpretation.

2.1 Component 1: Exact Pointwise RMSE-Delta Contribution

Define

$$c_i = \frac{(e_i^{la})^2 - (e_i^a)^2}{n(R_{la} + R_a)}, \quad R_{la} + R_a > 0.$$

Then

$$\Delta_{\text{RMSE}} = \sum_{i=1}^n c_i.$$

Thus c_i is an exact additive pointwise contribution to the RMSE difference. If $c_i < 0$, point i helps local-auto. If $c_i > 0$, point i hurts local-auto.

This is the cleanest answer to the vague phrase “the component of the dataset-level Truth-RMSE delta.” It is also the quantity used in most figures below.

For FB01, the dimension-one points slightly help local-auto in aggregate, but the dimension-two and dimension-three points hurt it in aggregate. This already suggests that the worse global Truth RMSE for local-auto is not simply caused by all dimension changes. The direction and magnitude depend on which local dimension was selected and where.

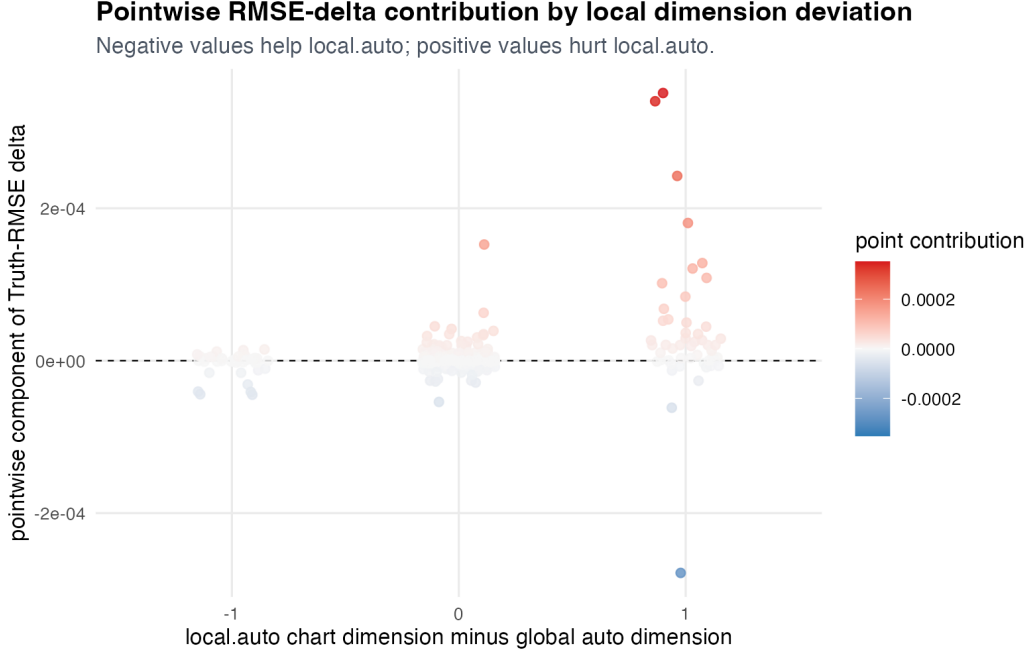


Figure 1: Pointwise RMSE-delta contributions by dimension deviation for FB01. The global-auto chart dimension is $d^a = 2$. The horizontal axis is $q_i = d_i^{la} - d^a$. Negative vertical values are points where local-auto improves Truth RMSE relative to auto; positive values are points where local-auto worsens Truth RMSE.

2.2 Component 2: Pointwise Error-Pair Geometry

The exact component c_i is additive, but it hides the sign of the raw errors. A second diagnostic is the error-pair plot:

$$x_i = e_i^a, \quad y_i = e_i^{la}.$$

The point i is improved by local-auto exactly when

$$|e_i^{la}| < |e_i^a|.$$

Equivalently, the point lies closer to the horizontal axis in the local-auto coordinate than in the auto coordinate. The lines

$$y = x \quad \text{and} \quad y = -x$$

mark the boundaries where the two squared errors are equal.

This plot asks a different question from Figure 1. Instead of asking how much each point contributes to the final delta, it asks whether local-auto changes the *error direction and magnitude*. It can reveal systematic bias changes, for example a model that tends to overshoot in one chart-dimension regime and undershoot in another.

2.3 Component 3: Spatial or Geometric Localization

A third interpretation is geometric. The same exact contribution c_i can be displayed over a representation of the observed state space. For FB01, the observed coordinate matrix is the four-component VALENCIA depth-1 dCST relative-abundance matrix. The figure below uses a two-dimensional PCA view of that four-dimensional coordinate matrix.

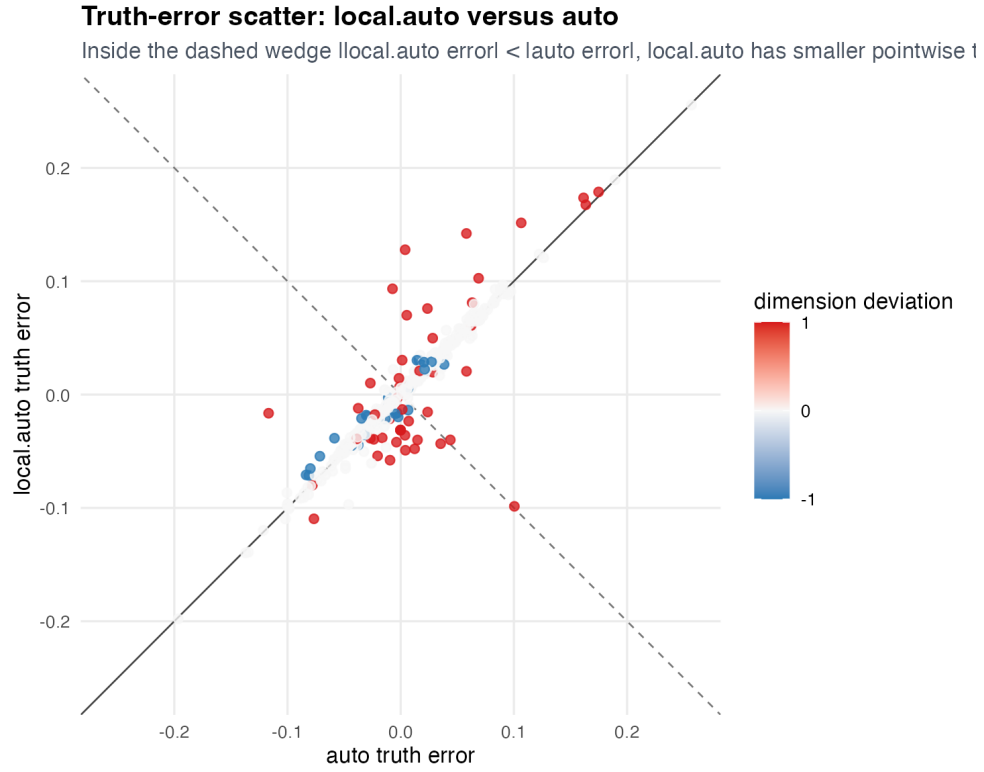


Figure 2: Truth-error scatter for FB01. The horizontal coordinate is the auto truth error e_i^a , and the vertical coordinate is the local-auto truth error e_i^{la} . Points between the two diagonal boundary lines have smaller absolute truth error under local-auto than under auto. Color shows the local-auto dimension deviation from the global-auto dimension.

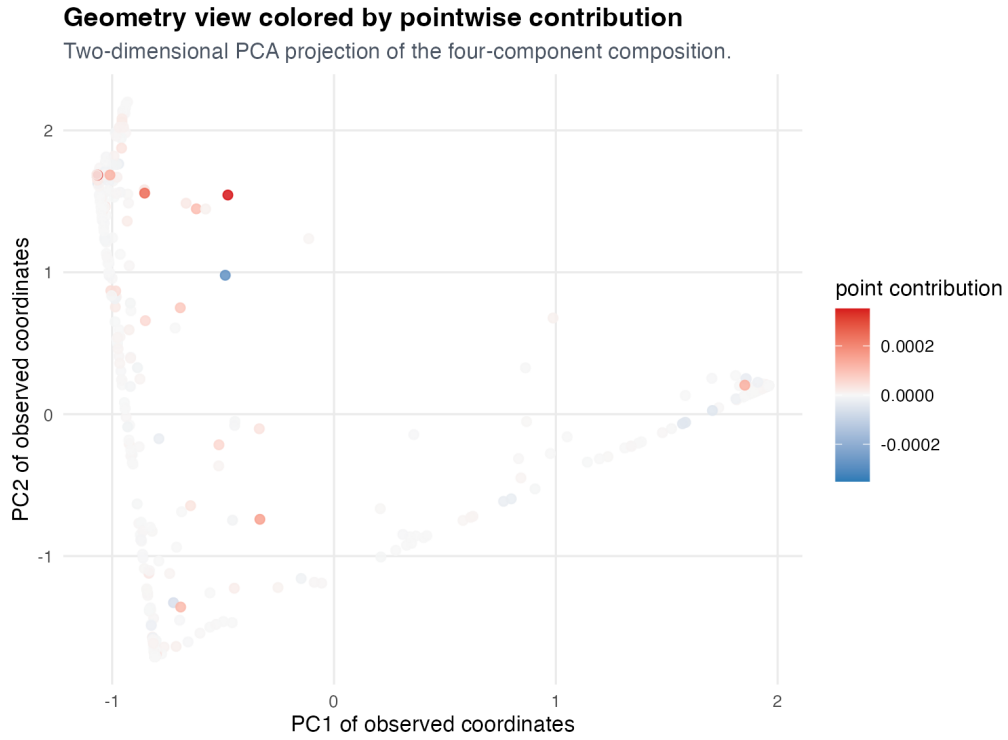


Figure 3: Geometry view for FB01, colored by the exact pointwise RMSE-delta contribution c_i . Blue points help local-auto, and red points hurt local-auto. The plot uses the first two principal components of the observed four-component relative-abundance coordinates, only as a visualization device.

This view answers whether the delta is spatially localized. If red and blue regions form coherent patches, then the local-dimension rule is interacting with geometry. If the colors are scattered without structure, then the difference may be more noise-like or driven by individual leverage points.

2.4 Component 4: Cumulative Influence or Leverage

A fourth interpretation asks whether the dataset-level delta is concentrated in a small number of points. Sort points by decreasing absolute contribution:

$$|c_{(1)}| \geq |c_{(2)}| \geq \cdots \geq |c_{(n)}|.$$

Then plot

$$S_m = \sum_{r=1}^m c_{(r)}, \quad m = 1, \dots, n.$$

If the curve reaches most of the final delta quickly, a small number of points dominate the comparison.

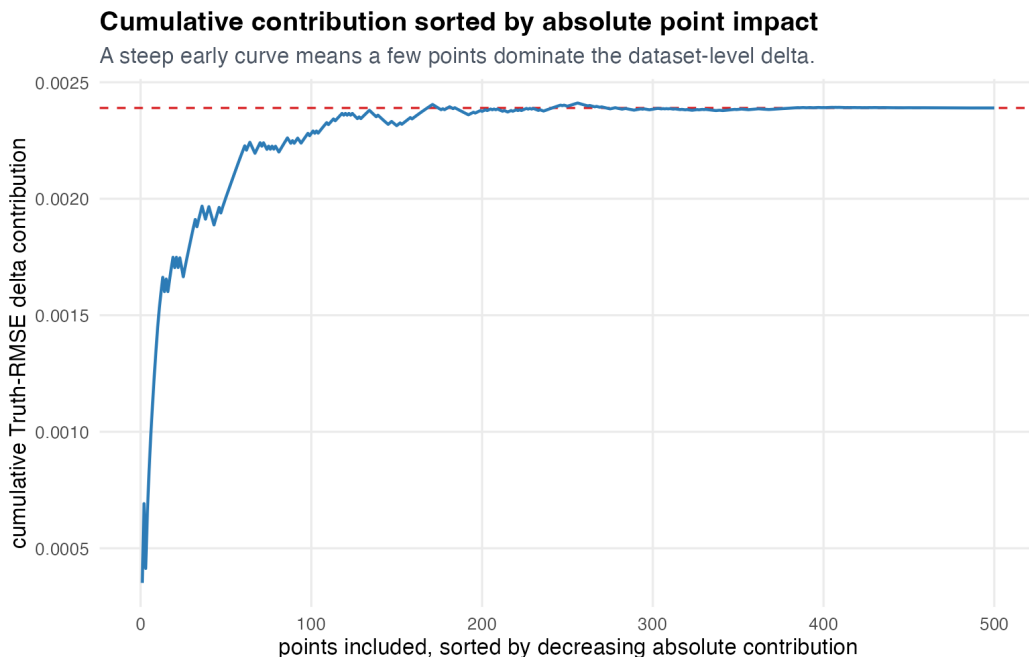


Figure 4: Cumulative contribution for FB01 after sorting points by decreasing absolute pointwise contribution. The dashed horizontal line is the final dataset-level Truth-RMSE delta. A steep initial curve indicates that a small subset of points has high leverage for the local-auto versus auto comparison.

For FB01, the ten largest absolute point contributions account for approximately 60.60% of the final signed delta, and the twenty largest account for approximately 71.32%. This suggests that the observed difference is not purely diffuse. A relatively small collection of points contributes substantially to the final comparison.

2.5 Component 5: Grouped or Regional Contribution

Finally, pointwise components can be aggregated over a group g . Let G_g be a subset of point indices, for example all points with a given local dimension, all points in a biological component label, or all

points in a synthetic stratum. Define

$$C_g = \sum_{i \in G_g} c_i.$$

The group contributions add to the same global delta:

$$\Delta_{\text{RMSE}} = \sum_g C_g,$$

provided the groups form a partition of the dataset.

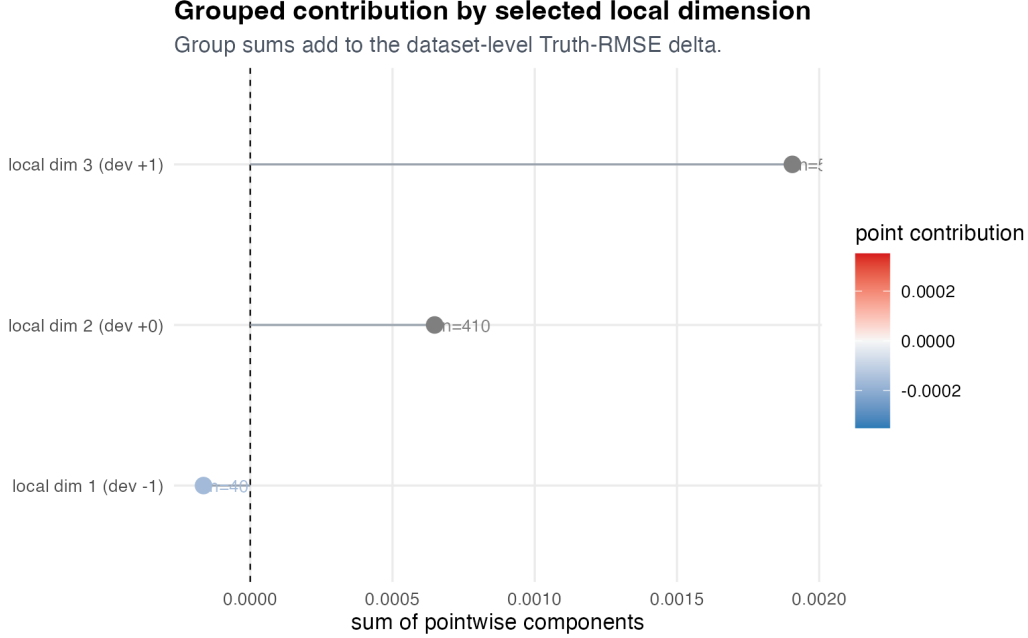


Figure 5: Grouped contribution by local-auto selected dimension for FB01. Each point is the sum of exact pointwise contributions over all data points with a given local dimension. Group sums add to the full Truth-RMSE delta.

The grouped decomposition for FB01 is:

$$\begin{aligned} d_i^{la} = 1 & : -0.000164 \quad (40 \text{ points}), \\ d_i^{la} = 2 & : 0.000648 \quad (410 \text{ points}), \\ d_i^{la} = 3 & : 0.001906 \quad (50 \text{ points}). \end{aligned}$$

Thus, in this example, points where local-auto reduced dimension below the global-auto dimension helped slightly. Points where local-auto matched the global dimension hurt modestly, and points where local-auto increased dimension to three hurt most in aggregate.

3 Interpretation for FB01

The five diagnostics give a coherent first read of FB01. The global-auto model used a single chart dimension of 2. The local-auto model mostly agreed with this dimension, but it selected dimension one at 40 points and dimension three at 50 points. The net result was a small worsening of Truth RMSE:

$$R_{la} - R_a = 0.0023897.$$

The most informative feature is not simply that local-auto changed dimensions. Rather, different changes had different consequences. Dimension-one selections were slightly beneficial in aggregate. Dimension-three selections were unfavorable in aggregate. This pattern suggests a useful next diagnostic: inspect whether the dimension-three points correspond to genuinely higher-dimensional local neighborhoods or to unstable local dimension overestimation near sparse, noisy, or boundary-heavy regions.

The cumulative contribution plot also says that the comparison is somewhat high-leverage. A small number of points explains a large part of the signed delta. Therefore, before concluding that local-auto is globally worse on this kind of dataset, it would be useful to identify these points in the original coordinate space and examine their local neighborhoods, selected supports, and fold behavior.

4 Recommended Use in the HTML Report

The HTML report should not include every diagnostic for every dataset at full detail, because that would become visually heavy. A practical compromise is:

1. Define the exact contribution c_i once.
2. For each dataset, show the contribution-versus-dimension plot and the grouped contribution summary.
3. Add the error-error scatter and geometry-colored contribution view for selected datasets where the grouped summary indicates a notable difference.
4. Keep the full pointwise CSV as a linked audit artifact rather than a large table in the report body.

This would make the HTML report more explanatory without turning it into a full per-point diagnostic atlas.

Conclusion

The exact pointwise decomposition makes the dataset-level Truth-RMSE delta auditable. It converts the question “which model won?” into more precise questions: which points helped, which hurt, whether those points form geometric regions, whether a few points dominate, and whether grouped local-dimension choices explain the result. For FB01, local-auto is slightly worse overall, but the result is structured: dimension-one local choices help slightly, dimension-three choices hurt most in aggregate, and a relatively small number of points contributes strongly to the final delta.

A Proof of the Pointwise Truth-RMSE Delta Decomposition

Assume

$$R_{la} + R_a > 0.$$

If $R_{la} = R_a = 0$, then both models recover the truth perfectly and $\Delta_{\text{RMSE}} = 0$, so no nontrivial decomposition is needed.

Starting from the definition,

$$\Delta_{\text{RMSE}} = R_{la} - R_a.$$

Use the elementary identity

$$x - y = \frac{x^2 - y^2}{x + y}, \quad x + y > 0.$$

With $x = R_{la}$ and $y = R_a$,

$$R_{la} - R_a = \frac{R_{la}^2 - R_a^2}{R_{la} + R_a}.$$

By definition of Truth RMSE,

$$R_{la}^2 = \frac{1}{n} \sum_{i=1}^n (e_i^{la})^2, \quad R_a^2 = \frac{1}{n} \sum_{i=1}^n (e_i^a)^2.$$

Therefore

$$R_{la}^2 - R_a^2 = \frac{1}{n} \sum_{i=1}^n \left\{ (e_i^{la})^2 - (e_i^a)^2 \right\}.$$

Substituting this into the previous display gives

$$R_{la} - R_a = \sum_{i=1}^n \frac{(e_i^{la})^2 - (e_i^a)^2}{n(R_{la} + R_a)}.$$

Thus, if

$$c_i = \frac{(e_i^{la})^2 - (e_i^a)^2}{n(R_{la} + R_a)},$$

then

$$\Delta_{\text{RMSE}} = R_{la} - R_a = \sum_{i=1}^n c_i.$$

B Squared-RMSE Alternative

Sometimes it is simpler to decompose squared RMSE instead of RMSE. Define

$$b_i = \frac{(e_i^{la})^2 - (e_i^a)^2}{n}.$$

Then

$$R_{la}^2 - R_a^2 = \sum_{i=1}^n b_i.$$

This squared-error decomposition is more direct algebraically. The c_i decomposition used in the main text is just the squared-error decomposition rescaled by the positive factor $R_{la} + R_a$, so that the components sum to the actual RMSE delta rather than the MSE delta.